



NITI Aayog

Strategy for

New India@75

STRATEGY FOR NEW INDIA @ 75

MODULE - 5

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DOUBLING FARMERS' INCOME (I): MODERNIZING AGRICULTURE



OBJECTIVES



- Modernize agricultural technology, increase productivity, efficiency and crop diversification.
- Generate income and employment through a paradigm shift that ensures food security, while maximizing value addition in agriculture.

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DOUBLING FARMERS' INCOME (I): MODERNIZING AGRICULTURE



PRESENT SCENARIO



- The existing yield levels of majority of crops remains much lower than the world average.
- The reasons are
 - Low irrigation
 - Low quality seeds
 - Low adoption of improved technology
 - Knowledge deficit about improved agricultural practices
- Close to 53% of cropped area is water stressed.
- Rainwater management practices are resource starved.
- This limits farmer's capacity to undertake multiple cropping and leads to inefficient utilization of land resources.

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DOUBLING FARMERS' INCOME (I): MODERNIZING AGRICULTURE



PRESENT SCENARIO



LARGE YIELD GAPS

- There is a gap between best scientific practices and best field practices.
- There is also a gap between best field practices and the average farmer.
- There exist significant yield gaps both amongst and within the States.
- Yield gaps are found to exist in even highly productive States like Punjab.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****PRESENT SCENARIO****77% WITH STAPLE CROPS**

Demand side factors favour the expansion of area under fruits and vegetables and livestock products.



- Staple crops like cereals, pulses and oilseeds occupy 77% of the total Gross Cropped Area (GCA).
- High Value Crops contribute an almost similar amount to total output, but, they occupy only 19% of the Gross Cropped Area.

Diversification to fruits and vegetables is likely to benefit small and medium farmers more than the larger ones.

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STAPLE CROPS



CEREALS



PULSES



OILSEEDS



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DOUBLING FARMERS' INCOME (I): MODERNIZING AGRICULTURE

CONSTRAINTS

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Huge gap exists between demand and supply of skills in agriculture and this hinders diversification, adoption of precision agriculture and on farm post-harvest value addition.

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Multiple private and public sources supplying different information to farmers create confusion.

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Agriculture Research is constrained by resource inadequacy, regulations and IPRs.



MAJOR BOTTLENECKS!



1

Use of outdated and inappropriate technology.

2

Predominance of small and marginal farmers at around 86%. Hence, affordability becomes a significant constraint.

3

On-farm adoption of technology is not taking place.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****INCREASE GROSS CROPPED AREA UNDER
IRRIGATION AND ADOPT IMPROVED SEEDS**

- Irrigation coverage needs to be increased to 53% of Gross Cropped Area by 2022-23.
- Focus should be on increasing coverage through micro irrigation.
- Dynamic Seed Development plans are required.
- Phase out old varieties of seeds and replace them with hybrid and improved seeds to enhance productivity.
- ICAR along with State Agricultural Universities should develop climate resilient varieties suitable for 128 agro climatic zones of the country.
- Strengthen seed testing laboratories.

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DOUBLING FARMERS' INCOME (I): MODERNIZING AGRICULTURE



WAY FORWARD

SOIL HEALTH CARDS SHOULD BE THE FULCRUM



- Strengthen the SHC scheme and include not nine, but all 16 parameters in the tests.
- This will ensure SHC based fertilizer distribution at the ground level.
- Seed SHCs with the Integrated Fertilizer Management System.
- Link SHCs with Kisan Credit cards and make SHCs mandatory for subsidies.
- Ensure proper functioning of the SHC labs.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****FERTILIZER SUBSIDY & PESTICIDE
REGULATORY FRAMEWORK****Government of India
Department of Fertilizers**

- The current lopsided fertilizer subsidy policy needs to bring secondary and micronutrients on the same nutrient based subsidy platform as phosphorus and potash.
- Align the pesticide regulatory framework with food safety laws to make adoption broad based.
- Targeted subsidy should be provided on liquid fertilizers to encourage fertigation with micro-irrigation.
- Strengthen extension activities to ensure that best practices reach the average farmer.



WAY FORWARD

MACRO-NUTRIENTS

- Macronutrients are divided into two classes i.e.,
 - Primary**
Nitrogen (N), phosphorus (P), and potassium (K), all are required in relatively large quantities by plants.
 - Secondary**
Calcium (Ca), magnesium (Mg), and sulphur (S), are required in lesser quantities relative to the primary category.

MICRO-NUTRIENTS

- Micronutrients are required by plants in far smaller quantities than any macronutrient.
 - Micronutrients**
They include zinc (Zn), iron (Fe), manganese (Mn), copper (Cu), boron (B), molybdenum (Mo), and chlorine (Cl). Relative to the macronutrients, micronutrients are required in trace quantities.



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DOUBLING FARMERS' INCOME (I): MODERNIZING AGRICULTURE



WAY FORWARD

CUSTOM HIRING CENTERS AND INVESTMENT SUBSIDIES



- Madhya Pradesh has had demonstrable success with their custom hiring center model.
- This hastens the pace of farm mechanization.
- The model should be replicated nationwide by employing rural youth and promoting entrepreneurship.
- Rather than power and water subsidies, investment subsidies for micro-irrigation can be provided through the DBT mode.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****STRENGTHEN EXTENSION SYSTEMS**

- Ensure synergy between Agriculture Technology Management Agency and Krishi Vigyan Kendras.
- ATMA should be reoriented to include bottom up planning at district and block levels to develop Strategic Research Extension Plans.
- Promotion of Public Private Partnership Model in the extension delivery.
- Each KVK in possession of approximately 50 acres of land should incubate private sector initiatives in extension delivery.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****SUSTAINABLE WATER USE IN AGRICULTURE**

- About 83% of water is used in agriculture.
- The solution to resolve India's water crisis lies in conserving water in agriculture.
- Therefore, more efficient irrigation technologies, water harvesting and better crop selection must be encouraged.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****HIGH VALUE CROPS**

- The area under fruits and vegetables needs to increase by 5% every year.
- Farmers should be weaned away from cereal crops to HVCs.
- Regional production belts for high value crops should be identified.
- Only 10% is under hybrids in vegetables.
- Shifting to hybrids has the potential to increase yields by 1.5 to 3 times and provide significant increase in income.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****ROOTSTOCK TECHNOLOGY FOR FRUITS**

- Rootstock technology can double the production of fruits.
- It is also resilient to climate stress.
- Measures should be taken to standardize and promote usage of rootstocks to produce fruits.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****PROMOTE SMART HORTICULTURE**

- There are some pockets, which are using the techniques such as high-density plantation, protected cultivation and organic production.
- These are to be documented at the national level and replication should be done as far as possible.
- Recycling and utilizing agricultural waste would give a fillip to farmers income.
- Strengthen market for organic products, branding plays a vital role.

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**DOUBLING FARMERS' INCOME (I):
MODERNIZING AGRICULTURE****WAY FORWARD****LIVESTOCK AND FISHERIES**

Integrate the Blue Revolution Scheme with Mahatma Gandhi NREGA and funds created through Mahatma Gandhi NREGA should be used to promote aquaculture and can be used to create potential clusters as well.

- Breed indigenous cattle with exotic breeds.
- An exotic species is one which is introduced into an ecosystem from outside. This usually involves human intervention.
- These can significantly increase milk productivity through the supply of cattle with enhanced milk potential.
- Installing of bulk milk chillers and facilities for high value conversion of milk are needed to promote dairy in the States.
- Private sector should be incentivized to create a value chain for dairy products at the village level.

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